

Phase 35: Sequence Filter Operator

The sequence filter operator restricts a sequence to elements from a specified set. It removes elements not in the filter set while preserving order.

ASCII notation: $s \upharpoonright A$ Unicode alternative: $s \restriction A$ (U+21BE)

Example 1 : Basic Filter Operation

Filter a sequence to only include elements from a set.

$$\begin{array}{l} s1 : \text{seq } \mathbb{N} \\ \text{evens1} : \mathbb{P } \mathbb{N} \\ \text{filtered1} : \text{seq } \mathbb{N} \\ \hline s1 = \langle 1, 2, 3, 4, 5, 6 \rangle \wedge \text{evens1} = \{2, 4, 6, 8\} \wedge \text{filtered1} = (s1 \restriction \text{evens1}) \end{array}$$

The filtered sequence is $\langle 2, 4, 6 \rangle$, keeping only elements that are in evens and preserving their original order.

Example 2 : Filter with Set Comprehension

$[Title2, Length2]$

$$\begin{array}{l} \text{all_titles} : \text{seq } Title2 \\ \text{long_titles} : \mathbb{P } Title2 \\ \text{filtered_titles} : \text{seq } Title2 \\ \hline \text{long_titles} = \{ t : Title2 \mid true \} \wedge \text{filtered_titles} = \text{all_titles} \restriction \text{long_titles} \end{array}$$

This filters a sequence of titles to include only titles with length greater than 120 minutes.

Example 3 : Comparison with Range Restriction

Sequence filter (filter) is different from range restriction ($\triangleright$): - filter: operates on sequences, preserves order - $\triangleright$: operates on relations, restricts range

$$\begin{array}{l} s3 : \text{seq } \mathbb{N} \\ R3 : \mathbb{N} \leftrightarrow \mathbb{N} \\ A3 : \mathbb{P } \mathbb{N} \\ \text{filtered_seq3} : \text{seq } \mathbb{N} \\ \text{restricted_rel3} : \mathbb{N} \leftrightarrow \mathbb{N} \\ \hline s3 = \langle 1, 2, 3, 4, 5 \rangle \wedge \\ R3 = \{1 \mapsto 10, 2 \mapsto 20, 3 \mapsto 30\} \wedge \\ A3 = \{2, 3, 4\} \wedge \\ \text{filtered_seq3} = s3 \restriction A3 \wedge \\ \text{restricted_rel3} = R3 \triangleright \{10, 20\} \end{array}$$

filtered_seq is $\langle 2, 3, 4 \rangle$ (sequence elements in A) restricted_rel is $\{1 \mapsto 10, 2 \mapsto 20\}$ (relation pairs with range in $\{10, 20\}$)

Example 4 : Practical Example - Video Database

$[Title4, Length4]$

$Rating4 ::= G4 \mid PG4 \mid PG13_4 \mid R4$

$catalog4 : seq (Title4 \times Length4 \times Rating4)$ $family_friendly4 : \mathbb{P} Rating4$
$family_friendly4 = \{G4, PG4\}$

The family_friendly set contains ratings G and PG. A filter could be applied to the catalog sequence to only include entries with ratings in this set.

Example 5 : Multiple Filters

$s5 : seq \mathbb{N}$ $evens5 : \mathbb{P} \mathbb{N}$ $large5 : \mathbb{P} \mathbb{N}$ $result5 : seq \mathbb{N}$
$s5 = \langle 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \rangle \wedge$ $evens5 = \{2, 4, 6, 8, 10, 12\} \wedge$ $large5 = \{5, 6, 7, 8, 9, 10, 11\} \wedge$ $result5 = (s5 \upharpoonright evens5) \upharpoonright large5$

Multiple filters can be applied sequentially. First filter to evens gives $\langle 2, 4, 6, 8, 10 \rangle$, then filter to large gives $\langle 6, 8, 10 \rangle$.

Example 6 : Empty Filter

$s6 : seq \mathbb{N}$ $empty_set6 : \mathbb{P} \mathbb{N}$ $disjoint_set6 : \mathbb{P} \mathbb{N}$ $result6a : seq \mathbb{N}$ $result6b : seq \mathbb{N}$
$s6 = \langle 1, 2, 3 \rangle \wedge$ $empty_set6 = \{\} \wedge$ $disjoint_set6 = \{10, 20, 30\} \wedge$ $result6a = s6 \upharpoonright empty_set6 \wedge$ $result6b = s6 \upharpoonright disjoint_set6$

Filtering by an empty set or disjoint set results in an empty sequence. $result1 = \langle \rangle \wedge result2 = \langle \rangle$

Example 7 : Bag Union (Related Phase 35 Feature)

Phase 35 also includes the bag union operator. ASCII notation: $b1 \uplus b2$ Unicode alternative: $b1 \uplus b2$ (U+228E)

$b1 : bag \mathbb{N}$ $b2 : bag \mathbb{N}$ $combined : bag \mathbb{N}$
$b1 = \llbracket 1, 1, 2, 3 \rrbracket \wedge b2 = \llbracket 2, 3, 3, 4 \rrbracket \wedge combined = b1 \uplus b2$

Bag union adds multiplicities: $combined = \llbracket 1, 1, 2, 2, 3, 3, 3, 4 \rrbracket$ Element 1 appears 2 times (2 from b1), element 2 appears 2 times (1 from each), element 3 appears 3 times (1 from b1, 2 from b2), element 4 appears 1 time (1 from b2).